

ModeSplit Software User Guide

1. Definitions

Cost: Direct incurs costs user incurs in his daily regular trips such as parking, gas, and tolls.

Time: Overall time consumed in daily regular trips for waiting, parking, and access time separately; but, except for walking time.

FRICT: is 0.5 cost of parking + walking from parking 6 cents/min.

WALK PLS is walking time in sec.

CAR: is car ownership.

RUSH: it is time of day peak/off peak.

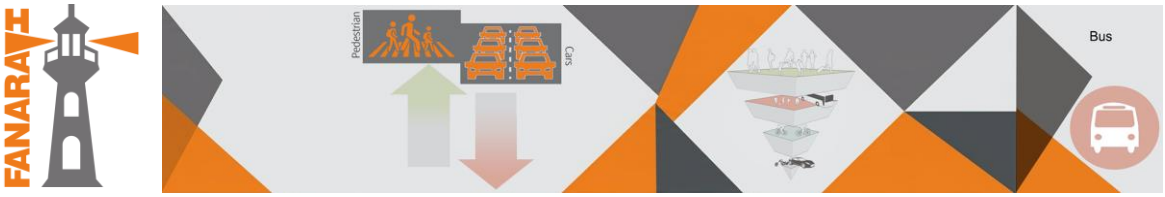
Driven mode: Refers to rides where you are passenger rather than the driver (e.g., rideshare apps like Uber, taxis, or hired private cars).

2. Introduction

Modal split models assign trips to various alternative travel modes. It is the most important classic models in transport planning. It is the key role played by public transport in policy making. Transport modes choice process is crucial also in urban road planning where this process can provide a perspective about the roads configuration epically after Public Transport received that great attention. In most cases, Public Transport make use of road space more efficiently than other modes . Also they have more social benefits like if more people begin to use public transport, there will be less congestion on the roads and the accidents will be less. Therefore, mode choice model can utilize resources, maximize benefits, and manage infrastructure spending more efficiently.

Here, Fanarah Team has proposed this open-source contribution (Fanarah Mode Split) (version 1) as implementation side for Modal Split model assign adjusted by Paul Inglis and his colleagues for short trips among travel mode alternatives.

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3. Run the model

Fanarah Mode Split Software has two paths (figure 1).

First, use aggregation level by selecting "Aggregate" tab. Fill the variables values in left side by selecting mode from the combo list box. Enter the variable values for every single mode and notice that not all variables are the same in terms of transport mode. Variable text boxes are related to the significant variables depending on the research model. Figure 2 shows the Aggregate tab functions.

Second, selecting "DisAggregate" tab to use Disaggregate data. Select the CSV table file directory from your computer after checking the schema table (table 1).

Table 1 – Data table schema

Attribute	DataType	Domain	Unique
CaseNumber	Integer	1 to 30	No
mode	Integer	1 to 4	No
Cost	float		No
TravelTime	float	1 to 600	No
Frict	float		No
WalkPlus	float		No
Car	float	1 – 2 – 3 – 4	No
Rush	float	0 – 1	No
Age	float		No

CSV is a text based format stands for Comma Separated Values, user can open and edit it with any text editor or EXCEL to have a .csv

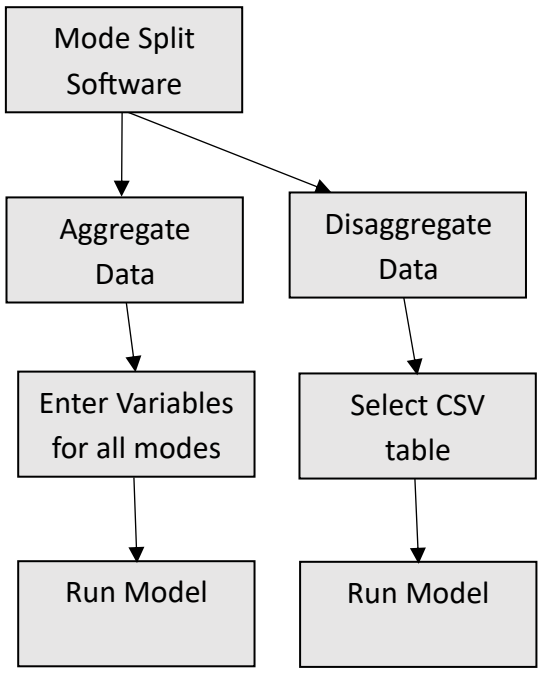
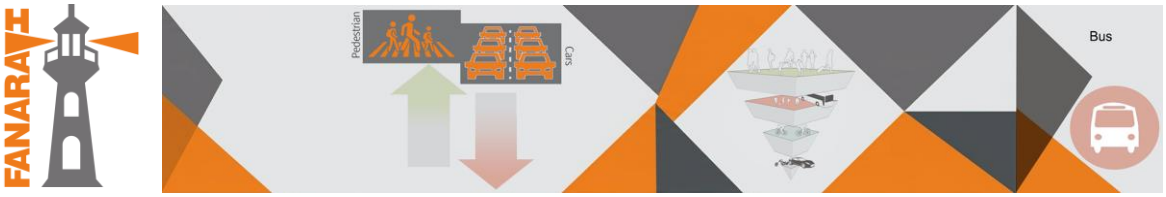


Figure 1 – software process

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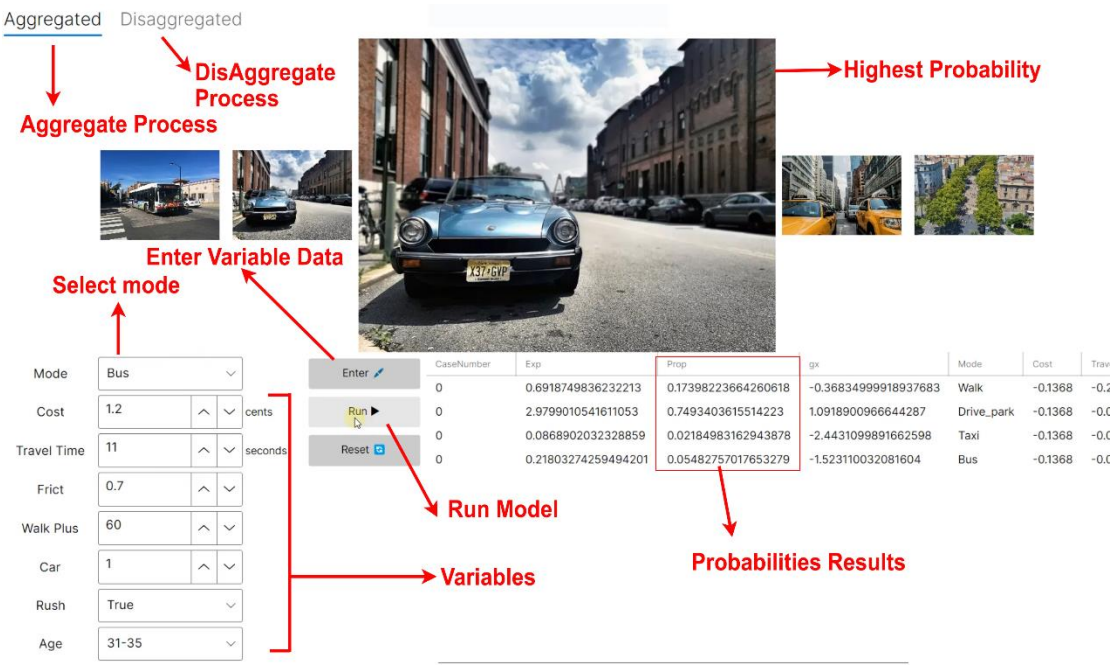


Figure 2 – Aggregate tab functions

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